Blueprint

The Clock That Never Ticks: a machine-checked path from the dilogarithm to quantum speed limits in Lean 4

Special functions · the dilogarithm and the Clausen function

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Statements

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At a glance

What the paper contributes

A sorry-free Lean 4 path from the dilogarithm, which Mathlib cites but does not define, to a quantum state that never orthogonalizes.

- **The real dilogarithm** with Euler's reflection, Landen's transformation, the duplication formula and $\mathrm{Li}_2(1/2)$ (Theorem 1); the golden-ratio values and the Lee–Yang $c_{\mathrm{eff}} = 2/5$ from a 3×3 linear system, without the five-term relation.
- **The circle**: the Clausen function and Catalan's constant $G = \mathrm{Cl}_2(\pi/2)$; the Fejér–Jackson inequality (Theorem 2) and, by Abel summation without asymptotics, $\mathrm{Cl}_2(\theta) \geq \sin(\theta)/2$ on $(0, \pi)$.
- **Speed limits**: Margolus–Levitin and an L^1 form of Mandelstam–Tamm for a scalar spectral measure (Theorem 3).
- **The assembly** (Theorem 4): populations $6/(\pi^2 n^2)$ on levels $E_n = n$ give infinite mean energy and variance, so both bounds are silent, and yet the state never reaches an orthogonal state.
- The mathematics is classical; the paper claims the formalization and the assembly, and records the Eneström–Kakeya reading as an observation (Remark 1).

🚀 **Start here: Theorem 1**: Euler reflection, Landen, duplication and $\mathrm{Li}_2(1/2)$ for the real dilogarithm.

Main results:

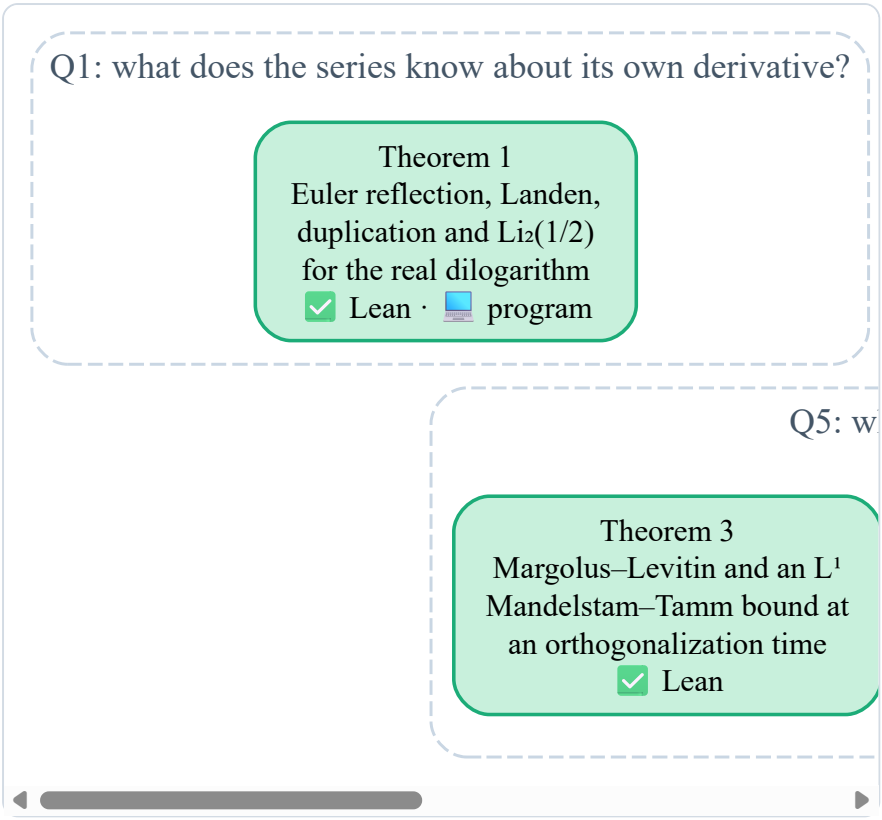
- **Theorem 4**: the weight-2 zeta state never reaches an orthogonal state

5 results | ✅ **5** entirely in Lean | ◐ **0** partly in Lean

💻 **3** checked by a program | written proof only (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 **Setting (copied from the paper, §1)**

The file `Mathlib/Analysis/SpecialFunctions/Integrals/LogTrigonometric.lean` of Mathlib … describes one of its integrals by referring to the dilogarithm

$$\mathrm{Li}_2(x) \;=\; \sum_{n\ge 1} \frac{x^n}{n^2}.$$

(§3) In the Rogers normalization $\mathrm{L}(x) = \mathrm{Li}_2(x) + \frac{1}{2}\ln x \ln(1-x)$ the $\ln^2$ defects cancel.
(§4) On the circle the dilogarithm splits into a tame half and a wild half:

$$\mathrm{Li}_2(e^{i\theta}) \;=\; \underbrace{\sum_{n\ge 1} \frac{\cos n\theta}{n^2}}_{=\,\frac{\pi^2}{6} - \frac{\pi\theta}{2} + \frac{\theta^2}{4}} \;+\; i \underbrace{\sum_{n\ge 1} \frac{\sin n\theta}{n^2}}_{=\,\mathrm{Cl}_2(\theta)} \qquad (\theta \in [0, 2\pi]).$$

(§6) A state $|\psi_0\rangle = \sum_n c_n |E_n\rangle$ evolves with autocorrelation $S(t) = \sum_n p_n e^{-iE_n t}$, $p_n = |c_n|^2$, and reaches an orthogonal state at time τ iff $S(\tau) = 0$. Take populations $p_n = \frac{6}{\pi^2 n^2}$ on levels $E_n = n$: the *weight-2 zeta state*.

What this blueprint shows.

The numbered statements of the paper, with their proofs, dependencies and what checked them, under the sections that hold them (with the paper's numbers), and the figures and tables they use. The full text is in the PDF of the paper.

2 Q1: what does the series know about its own derivative?

Theorem 1. Lean (Li₂_add_Li₂_one_sub, Li₂_landen, Li₂_sq, Li₂_one_half)✓

p. 3

On the stated domains,

$$\begin{aligned} \mathrm{Li}_2(x)+\mathrm{Li}_2(1-x) &= \tfrac{\pi^2}{6}-\ln x\,\ln(1-x) && x\in(0,1)\\ \mathrm{Li}_2(x)+\mathrm{Li}_2\!\left(\tfrac{x}{x-1}\right) &= -\tfrac{1}{2}\ln^2(1-x) && x\in(0,\tfrac{1}{2})\\ \mathrm{Li}_2(x^2) &= 2\big(\mathrm{Li}_2(x)+\mathrm{Li}_2(-x)\big) && |x|\le 1\\ \mathrm{Li}_2(\tfrac{1}{2}) &= \tfrac{\pi^2}{12}-\tfrac{\ln^2 2}{2}. \end{aligned}$$

Lean · ours show

added to the Lean repository after publication (21-oct-2026): L(1) = π²/6 and the Lee–Yang ratio c_eff = L(1/φ²)/L(1) = 2/5 as one statement (published: only L(1/φ²) = π²/15)

- `Dilog.Li₂_add_Li₂_one_sub`
- `Dilog.Li₂_landen`
- `Dilog.Li₂_sq`
- `Dilog.Li₂_one_half`
- `Dilog.Li₂_inv_goldenRatio_sq`
- `Dilog.Li₂_inv_goldenRatio`
- `Dilog.Li₂_neg_inv_goldenRatio`
- `Dilog.rogersL_inv_goldenRatio_sq`
- `Dilog.rogersL_gold_sum`
- `Dilog.rogersL_one`
- `Dilog.leeYang_effective_central_charge`
- `Dilog.Li₂_neg_one`

✅ Checked when this page was built: Basic.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — Li₂_add_Li₂_one_sub, Li₂_landen, Li₂_sq, Li₂_one_half, Li₂_inv_goldenRatio_sq, Li₂_inv_goldenRatio, Li₂_neg_inv_goldenRatio, rogersL_inv_goldenRatio_sq, rogersL_gold_sum, rogersL_one, leeYang_effective_central_charge, Li₂_neg_one: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean Basic.lean` (SHA-256 of the compiled file: `b7bc57fab92df8e3`…; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** classical (Euler 1768, Landen 1780; Lewin, Zagier); the paper claims, from its June 2026 checks, the first formalization of the real dilogarithm and these identities in a proof assistant (§7(i)).

↳ Not used by another result of the paper.

Proof

Formalized in Lean (see the linked declarations).

🛡️ **Evidence & scope**

- **Program:** $\mathrm{Li}_2(1/\varphi^2)$, $\mathrm{Li}_2(1/2)$, $\mathrm{Li}_2(1/\varphi)$, $\mathrm{Li}_2(-1/\varphi)$, $\mathrm{Li}_2(1) = \pi^2/6$ and $\mathrm{Li}_2(-1) = -\pi^2/12$ against the series (mpmath polylog), to 10^{-12}; rerun 21-oct: passes. `figures/dilog_fig1_golden.py (godsil-gutman-lean, the script of the paper, Appendix A)`

5 Q4: why is the wild half positive?

Theorem 2. Lean (`fejerSum_pos`)✓ p. 5

> *For every* $M \geq 1$ *and* $\theta \in (0, \pi)$: $\displaystyle\sum_{k=1}^{M} \frac{\sin k\theta}{k} > 0.$

Lean · ours show

added to the Lean repository after publication (21-oct-2026): the duplication $\mathrm{Cl}_2(2\theta) = 2\,\mathrm{Cl}_2(\theta) - 2\,\mathrm{Cl}_2(\pi - \theta)$ (the circle shadow of the duplication formula of Theorem 1), $\mathrm{Cl}_2(2\pi/3) = (2/3)\,\mathrm{Cl}_2(\pi/3)$, and $\mathrm{Li}_2(i) = -\pi^2/48 + iG$

- `Dilog.fejerSum_pos`
- `Dilog.dirichlet_telescope`
- `Dilog.sin_last_nonneg_at_localMin`
- `Dilog.sin_div_two_le_Cl₂`
- `Dilog.Cl₂_pos`
- `Dilog.Cl₂_pi_div_two`
- `Dilog.hasSum_cos_div_sq`
- `Dilog.Cl₂_two_mul`
- `Dilog.Cl₂_two_pi_div_three`
- `Dilog.Li₂c_I`

✅ Checked when this page was built: Clausen.lean, ComplexBoundary.lean, FejerJackson.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — fejerSum_pos, dirichlet_telescope, sin_last_nonneg_at_localMin, sin_div_two_le_Cl₂, Cl₂_pos, Cl₂_pi_div_two, hasSum_cos_div_sq, Cl₂_two_mul, Cl₂_two_pi_div_three, Li₂c_I: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean Clausen.lean ComplexBoundary.lean FejerJackson.lean` (SHA-256 of the compiled file: `78931a748edb7c0c…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** classical (Fejér 1910, Jackson 1911); the paper records two proof-engineering choices, the division-free Dirichlet telescope and the asymptotics-free bound $\mathrm{Cl}_2(\theta) \geq \sin(\theta)/2$ (§7(ii)).

↳ Used in: Theorem 4 (p. 7)

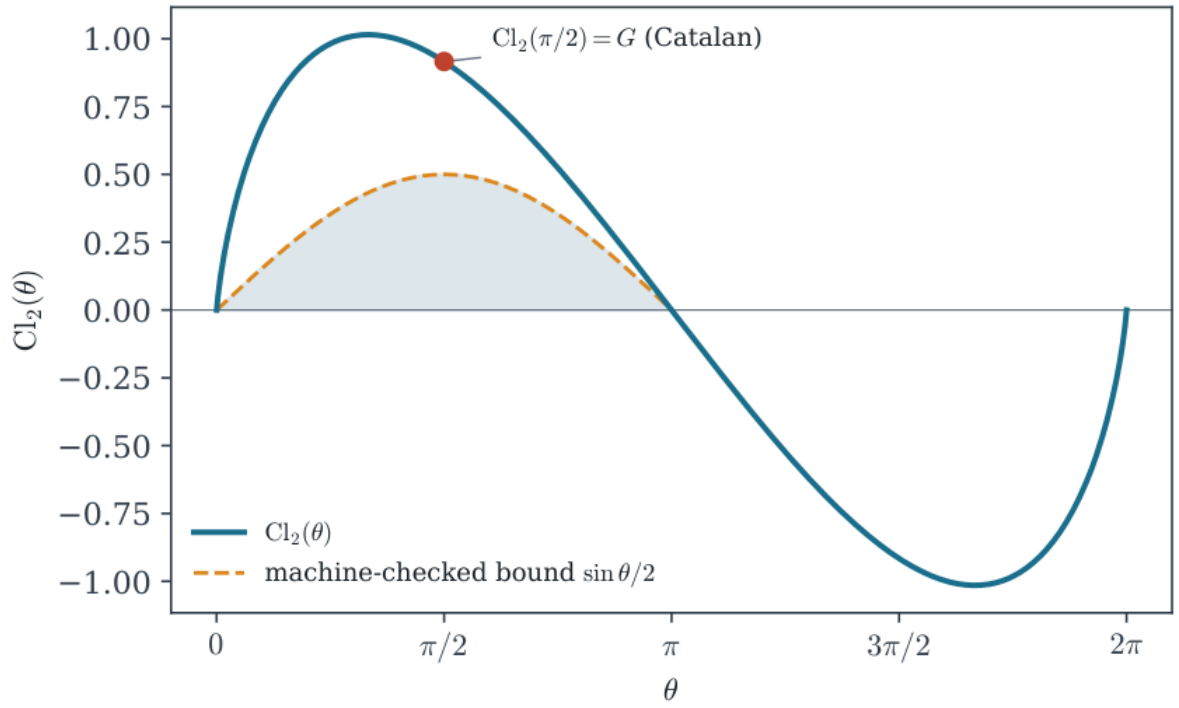
Proof

Formalized in Lean (see the linked declarations).

🛡️ **Evidence & scope**

- **Program:** $\mathrm{Cl}_2(\theta) \geq \sin(\theta)/2$ on 399 points of $(0, \pi)$, $\mathrm{Cl}_2(\pi/2)$ = Catalan's G, $\mathrm{Cl}_2(\pi) = 0$ and $\mathrm{Cl}_2(2\pi - \theta) = -\mathrm{Cl}_2(\theta)$ at three points, to 10^{-25} (mpmath); rerun 21-oct: passes. `figures/dilog_fig2_clausen.py (godsil-gutman-lean, the script of the paper, Appendix A)`

Figure 2



The Clausen function on $[0, 2\pi]$ with the machine-checked lower bound $\sin(\theta)/2$ (shaded) on $(0, \pi)$, and Catalan's constant at the quarter circle. The generating script asserts the bound on a 400-point grid at 30 digits.

6 Q5: what physical thing beats to this function?

Theorem 3. Lean (`margolus_levitin`, `mandelstam_tamm_L1`; $\hbar = 1$)✓ p. 7

> *If* $S(\tau) = 0$ *with* $\tau \geq 0$, *then*
>
> $$\pi \;\leq\; 2\,\langle E\rangle\,\tau \qquad \text{and} \qquad 1 \;\leq\; D_1\,\tau, \quad D_1 = \textstyle\sum_n p_n |E_n - \bar{E}|.$$

Lean · ours show

added to the Lean repository after publication (21-oct-2026): $D_1 \leq \Delta E$ (Cauchy–Schwarz) and the textbook weak Mandelstam–Tamm bound $\tau \geq 1/\Delta E$, which the paper derives in words (§6)

- `QSL.margolus_levitin`
- `QSL.one_sub_le_cos`
- `QSL.mandelstam_tamm_L1`
- `QSL.l1_dev_le_sqrt_variance`
- `QSL.mandelstam_tamm_weak`

✅ Checked when this page was built: MandelstamTamm.lean, MargolusLevitin.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — margolus_levitin, one_sub_le_cos, mandelstam_tamm_L1, l1_dev_le_sqrt_variance, mandelstam_tamm_weak: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean MandelstamTamm.lean MargolusLevitin.lean` (SHA-256 of the compiled file: `224c15dcafc7d45b`…; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** classical (Mandelstam–Tamm 1945, Margolus–Levitin 1998); the paper states them for a scalar spectral measure and calls the fully real statements a reusable choice (§7(ii)). The sharp $\pi/(2\Delta E)$ is left open (§8).

↳ Not used by another result of the paper.

Proof

Formalized in Lean (see the linked declarations).

Theorem 4. ✅ Lean (`zetaState_never_orthogonal`)✓ 📕 p. 7

For every $\theta \in (0, 2\pi)$, the real and imaginary parts of the weight-2 zeta-state autocorrelation are not both zero: the state never reaches an orthogonal state.

Lean · ours **show**

added to the Lean repository after publication (19-oct-2026): the complex dilogarithm Li₂c on the closed disk, with Re Li₂c(e^{iθ}) the parabola and Im Li₂c(e^{iθ}) = Cl₂(θ), so the theorem reads Li₂(e^{iθ}) ≠ 0 on (0, 2π) as the paper says it in words; and (21-oct-2026) the infinite mean energy and infinite second moment of the state, which the paper states in words, and the floor the paper certifies only numerically (a 719-point grid): ‖Li₂ z‖ ≥ π²/12 on the whole unit circle, so |S| ≥ 1/2 for every θ (Li₂(w) = w∫∫dx dy/(1 − xyw) on the unit square, Re (1 − tz)⁻¹ ≥ (1 + t)⁻¹, and r → 1⁻). We have not found this floor stated; its proof is elementary and it may well be folklore

- `Dilog.zetaState_never_orthogonal`
- `Dilog.zetaStateRe_pi`
- `Dilog.Cl₂_two_pi_sub`
- `Dilog.Li₂c_exp_re`
- `Dilog.Li₂c_exp_im`
- `Dilog.Li₂c_exp_ne_zero`
- `Dilog.zetaState_meanEnergy_not_summable`
- `Dilog.zetaState_secondMoment_not_summable`
- `Dilog.norm_Li₂c_ge`
- `Dilog.zetaState_norm_ge_half`
- `Dilog.Li₂c_eq_mul_integral`

Lean · Mathlib, not ours **show**

Mathlib's Fourier series of the Bernoulli polynomials, at k = 1: it gives the real part of S (the parabola), used at θ = π

hasSum_one_div_nat_pow_mul_cos · `Mathlib/NumberTheory/ZetaValues.lean`, line 356

```
theorem hasSum_one_div_nat_pow_mul_cos {k : ℕ} (hk : k ≠ 0) {x : ℝ} (hx : x ∈
Icc (0 : ℝ) 1) :
    HasSum (fun n : ℕ => 1 / (n : ℝ) ^ (2 * k) * Real.cos (2 * π * n * x))
      ((-1 : ℝ) ^ (k + 1) * (2 * π) ^ (2 * k) / 2 / (2 * k)! *
        (Polynomial.map (algebraMap ℚ ℝ) (Polynomial.bernoulli (2 * k))).eval
x)
```

Mathlib commit `f0b4c8150aa0`, as this blueprint was built.

✅ Checked when this page was built: CircleFloor.lean, Clausen.lean, ComplexBoundary.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — zetaState_never_orthogonal, zetaStateRe_pi, Cl₂_two_pi_sub, Li₂c_exp_re, Li₂c_exp_im, Li₂c_exp_ne_zero, zetaState_meanEnergy_not_summable, zetaState_secondMoment_not_summable, norm_Li₂c_ge, zetaState_norm_ge_half, Li₂c_eq_mul_integral: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean CircleFloor.lean Clausen.lean ComplexBoundary.lean` (SHA-256 of the compiled file: `9c291eed2052702a`…; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** the assembly is the paper's: «whose physical reading we have not seen stated, though we would not be surprised to find it folklore among polylogarithm or QSL specialists» (§7(iii)).

↳ Used in: 1

Proof

> *Proof (as in Lean).* On $(0, \pi)$ the imaginary part is $-\mathrm{Cl}_2(\theta) \neq 0$ by the Fejér–Jackson bound. At $\theta = \pi$ the real part is the parabola value $-\pi^2/12 \neq 0$ (it is $-\eta(2)$). On $(\pi, 2\pi)$, the reflection $\mathrm{Cl}_2(2\pi - \theta) = -\mathrm{Cl}_2(\theta)$ hands the sign back to the first case.

💡 **In plain words (commentary, not part of the paper)**

Put weight $p_n = 6/(\pi^2 n^2)$ on the energy levels $n = 1, 2, 3, \ldots$ and let the state evolve. It becomes orthogonal to where it started exactly when its autocorrelation $S(\theta) = \sum_n p_n e^{-in\theta}$ vanishes. This S is the dilogarithm on the unit circle: its real part is a parabola, and its imaginary part is minus the Clausen function. On $(0, \pi)$ the Clausen function is at least $\sin(\theta)/2$, which follows from the Fejér–Jackson inequality, so the imaginary part is never zero there. At $\theta = \pi$ the parabola equals $-\pi^2/12$. On $(\pi, 2\pi)$ the mirror $\mathrm{Cl}_2(2\pi - \theta) = -\mathrm{Cl}_2(\theta)$ brings back the first case. So S never vanishes. Meanwhile the mean energy and the energy spread of this state are both infinite, so the two textbook speed limits say nothing about it. Each piece is classical; what the paper adds is the assembly and its machine-checked proof.

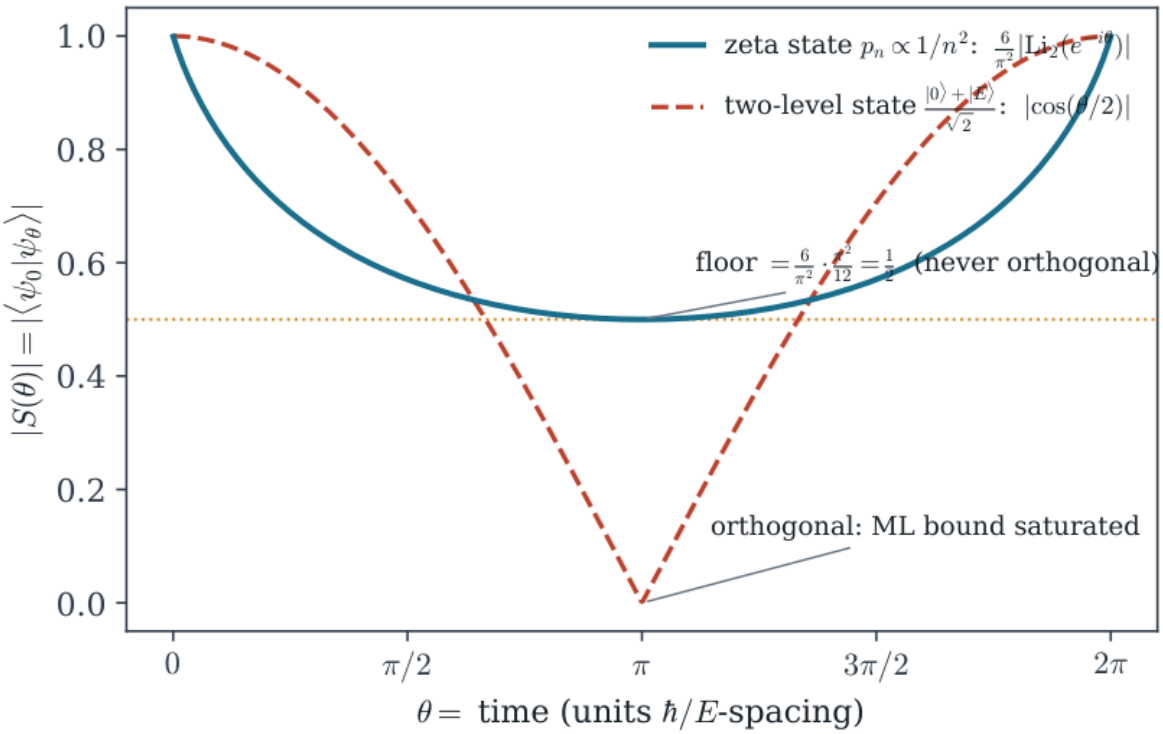
🛡️ **Evidence & scope**

- **Program:** min |Li₂(e^−iθ)| on a 719-point grid of (0, 2π) sits at θ = π and equals η(2) = π²/12, to 10⁻²⁰, so |S| ≥ 1/2 on that grid (mpmath); rerun 21-oct: passes. `figures/dilog_fig3_clock.py (godsil-gutman-lean, the script of the paper, Appendix A)`

Proof

> Formalized in Lean (see the linked declarations).

Figure 3



Two states with the same level spacing. The two-level state $(|0\rangle + |E\rangle)/\sqrt{2}$ (dashed) saturates Margolus–Levitin and strikes orthogonality at $\theta = \pi$. The zeta state (solid), with *infinite* mean energy, has $|S| \geq \frac{1}{2}$: the clock that never ticks. The script asserts the floor equals $\eta(2) = \pi^2/12$ at $\theta = \pi$.

References

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N. Hörnedal and O. Sönnerborn, The Margolus–Levitin quantum speed limit for an arbitrary fidelity, *Phys. Rev. Research* **5** (2023); see also the elementary proof in arXiv:2307.16854.

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L. Alha, Families of self-inverse functions and dilogarithm identities, arXiv:2509.07598.

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The mathlib Community, The Lean mathematical library, *CPP 2020*. find it